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Uncertainty, information, and disagreement of economic forecasters

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ABSTRACT
An information framework is proposed for studying uncertainty and disagreement of economic forecasters. This framework builds upon the mixture model of combining density forecasts through a systematic application of the information theory. The framework encompasses the measures used in the literature and leads to their generalizations. The focal measure is the Jensen–Shannon divergence of the mixture which admits Kullback–Leibler and mutual information representations. Illustrations include exploring the dynamics of the individual and aggregate uncertainty about the US inflation rate using the survey of professional forecasters (SPF). We show that the normalized entropy index corrects some of the distortions caused by changes of the design of the SPF over time. Bayesian hierarchical models are used to examine the association of the inflation uncertainty with the anticipated inflation and the dispersion of point forecasts. Implementation of the information framework based on the variance and Dirichlet model for capturing uncertainty about the probability distribution of the economic variable are briefly discussed.

KEYWORDS
Bayesian hierarchical model; entropy; Kullback-Leibler; mixture density; mutual information; survey of professional forecasters

JEL CLASSIFICATION
C11; C18; C41

1. Introduction

The approaches used for measuring economic uncertainty can be classified into three broad categories: survey-based, model-based, and using economic and financial indexes as proxies. Our main focus is on the surveys that solicit probability distributions from forecasters, but the information framework proposed here also is applicable to the model-based approach. We refer the reader to Jurado et al. (2015) and Bloom (2009) for the latest developments in model-based approach and the application of proxies, respectively.

Each quarter, the Federal Reserve Bank of Philadelphia, European Central Bank, the Bank of England, and some other central banks assess quarterly economic outlook through survey of economic forecasters. The respondents provide point forecasts and probability distributions for economic variables such as GDP, unemployment rate, and the GDP deflator for the current year and the following year. These databases have been important sources for studying uncertainty and disagreement about the economic variables by researchers and policymakers. Despite the success of these databases in attracting researchers to use forecasters' probability distributions for measuring uncertainty and disagreement, the following deficiencies in this area are evident.

- The finite mixture model for combining the forecast distributions has been proposed, however, there is no general framework which fully utilizes an important property of uncertainty functions for mixtures to measure uncertainty and disagreement.

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- Shannon entropy and Kullback–Leibler (KL) information have been considered for measuring uncertainty of economic forecasters and comparing forecast densities, however the full potential of information theory as a framework for measuring uncertainty and disagreement has not been realized.
- The dynamics of the uncertainty of the economic forecasters have been noted, but not formally studied in examining the relationships between uncertainty and other variables beyond some sporadic attempts.
- The uncertainty about solicited probabilities has been acknowledged but the uncertainty about the distribution of the economic variable has not been taken into account in measuring uncertainty of the economic forecasters.

The current state of measuring uncertainty and disagreement of economic forecasters and the approach of this paper can be best summarized by the following statement on the variance and entropy:

> "*The historical development has resulted in variance playing the central role in measuring dispersion, uncertainty, evaluating fit, and much more. This has made variance the implicit 'index' of choice in assessing and ranking random variables. But much axiomatic work points to other indices, particularly entropy classes, as superior measures of information, see Maasoumi (1993) for a survey. The simplicity of variance, however, remains a major attraction.*" (Ebrahimi et al., 1999)

This paper illustrates that a systematic application of information theory to the mixture model proposed by Wallis (2005) for combining density forecasts provides, a framework for measuring uncertainty and disagreement of economic forecasters.

The outline of the paper is as follows.

Section 2 gives an overview of measures used for uncertainty and disagreement of the economic forecasters. We will note that the variance remains the major attraction for measuring both uncertainty and disagreement, and sporadic attempts have been made to use entropy and KL information. We also note that while the mixture model has been proposed for combining density forecasts, it has not been utilized beyond the application of variance. The lack of a general framework for measuring uncertainty and disagreement is evident.

Section 3 presents the information framework for studying the notions of uncertainty, information, and disagreement about an economic variable. This framework builds upon the mixture model of Wallis (2005) by fully utilizing the important property of uncertainty functions for the mixture model, namely, any concave functional of the mixture density is larger than the mixture of the concave functional of the component densities. In the proposed framework, information is defined in terms of the uncertainty reduction and disagreement is defined in terms of divergence functions between probability distributions. The focal measure of this paper is the Jensen–Shannon (JS) divergence of the mixture. This measure bridges the use of entropy for measuring uncertainty based on survey of forecast distributions (Rich and Tracy, 2010; Soofi et al., 2009), the use of KL information (Bao et al., 2007; Hall and Mitchell, 2007; Mitchell and Hall, 2005; Mitchell and Wallis, 2010), and Sims's theory of rational inattention (Sims, 2003, 2006, 2011). The proposed framework illustrates the following assertion in a new context:

> "*Information theory provides a unification of known results, and leads to natural generalizations and derivation of new results* (Kullback, 1959, p. vii)".

Section 4 explores the dynamics of the uncertainty about the inflation rate over time using the survey of professional forecasters (SPF) database. The SPF survey solicits forecast distribution on a set of preassigned intervals (bins). The number of bins, interval widths, and the overall range have been changed several times since 1968, but no systematic approach is available for the adjustments of these changes. The information theory provides the normalized entropy index (Golan et al., 2000) for adjusting changes of the number of bins. In this section, we also develop Bayesian hierarchical models for the individual's and quarterly aggregate uncertainty about the inflation rate. We use these models to examine the association between the inflation rate uncertainty and point forecast, and the dispersion of point forecasts beyond the dynamic of uncertainty.

Section 5 presents moment-based information-theoretic uncertainty and disagreement measures. The forecast distributions collected by a survey are said to be noisy and their moments can be more reliable (So, 2013; D'Amico and Orphanides, 2008). The variance-based forecast distributions are derived by the maximum entropy (ME) principle. These measures are compared with the measures computed from the forecast distributions of the SPF data for the U.S. inflation rate, at the individual forecaster's level and at the quarterly aggregate level.

Section 6 briefly presents measuring uncertainty by taking into account the uncertainty about the probability distribution of the economic variable. It has been recognized that "it is difficult for forecasters to give precise numerical values for their subjective probabilities" Boero et al. (2014). The Dirichlet embedding (Soofi et al., 2009; Ebrahimi et al., 2011) combines the uncertainty about the probability distribution of the economic variable and the entropy of the forecaster's distribution. In Section 7 we summarize and discuss our findings and point out some extensions for future research.

2. Brief background

The SPF and similar surveys solicit point forecasts of an economic variable Y and forecast distributions in the form of estimates of probabilities that Y falls into a set of m preassigned intervals

$$p_k = \hat{P}(\xi_{k-1} \le Y \le \xi_k), \quad k = 1, \dots, m.$$

In the economics literature, various variance measures are used for quantifying uncertainty about Y based on these forecast distributions. The most basic measure is the variance σ_i^2 (or standard deviations) of the mid-interval points $\bar{\xi}_k = (\xi_{k-1} + \xi_k)/2$ of the forecaster distributions $\mathbf{p}_i = (p_{i1}, \dots, p_{im}), i = 1, \dots, n$ (Zarnowitz and Lambros, 1987; Lahiri et al., 1988). Giordani and Söderlind (2003) use the variance of the normal fit to the histogram, refer to the mid-interval method as a crude alternative for measuring variance of a histogram. These authors also discuss adding $1/12$ to the result of the mid-interval method, which in the statistics literature is known as the Sheppard's correction for calculating variance based on grouped data (Kendall and Stuart, 1962). Aggregate uncertainty measures are obtained by averaging the variances of the forecast distributions.

In an alternative approach, first the forecast distributions are aggregated in terms of their average,

$$\mathbf{p}_c = \frac{1}{n} \sum_{i=1}^{n} \mathbf{p}_i, \tag{1}$$

and then the uncertainty is measured by σ_c^2, computed similarly to the individual forecast distribution (Giordani and Söderlind, 2003; Wallis, 2005, 2008).

Considerable attention has been devoted to the following decomposition of the variance of the aggregate distribution:

$$\sigma_c^2 = \frac{1}{n} \sum_{i=1}^{n} \sigma_i^2 + \frac{1}{n} \sum_{i=1}^{n} (\mu_i - \mu_c)^2, \tag{2}$$

where μ_i is the mean of $\mathbf{p}_i$ and $\mu_c = \sum_{i=1}^{n} \mu_i/n$ is the mean of $\mathbf{p}_c$. Decomposition (2) is similar to the usual analysis of variance decomposition and Lahiri et al. (1988) interpreted (2) in terms of the familiar terminologies: the total variation, within respondents uncertainty, and between respondents uncertainty. Interpreting μ_i as the forecaster's point forecast, μ_c is the consensus (mean expected) forecast (Zarnowitz and Lambros, 1987; Lahiri et al., 1988), and the second term in (2) is a measure of disagreement (Giordani and Söderlind, 2003; Wallis, 2008). Rich and Tracy (2010) drew on the Boero et al.'s (2008) idea of "implied root mean subjective variance" and computed an analog for the average variance by subtraction, $\sigma_c^2 - \sigma_{\mathcal{F}}^2$, where $\sigma_{\mathcal{F}}^2$ is the variance of the point forecast $\mathcal{F}_i$ provided by the forecaster in addition to the forecast distribution. Assuming that $\mathcal{F}_i \approx \mu_i$, they called this measure "implied moment-based uncertainty"; for an excellent expository analysis of comparison of point forecasts and forecast distributions, see Engelberg et al. (2009).

Wallis (2005) proposed the following more general mixture model for combining density forecasts:

$$f_c(y) = \sum_{i=1}^{n} w_i f_i(y), \quad w_i > 0, \quad \sum_{i=1}^{n} w_i = 1. \tag{3}$$

However, this model is not used beyond explicating (1) and (2) in terms of the conditional variance. The mixture model formulation provides the basis for the information framework that unifies the measures considered in the literature.

The Shannon entropy has also been used for measuring uncertainty about economic variables based on survey data (Soofi et al., 2009; Rich and Tracy, 2010). Rich and Tracy (2010) recognized that the entropy of $\mathbf{p}_c$ decomposes into two terms as follows: "One of the terms would correspond to average uncertainty, but the other term would correspond to the dispersion in respondents' forecast uncertainty, not in their inflation forecasts." While they did recognize that the latter term is a KL divergence (private communication), they did not discuss this measure further. The expression for Shannon entropy of the normal distribution has been suggested as an adjustment for "comparing entropy-based and moment-based uncertainty measures" (Wallis, 2006) and as "the expected logarithmic score of the correct conditional density" (Mitchell and Wallis, 2010). The KL information has been used for motivating the likelihood ratio test of the fit of a forecast model to the "true" distribution (Bao et al. 2005; Mitchell and Hall, 2005; Hall and Mitchell, 2007; Mitchell and Wallis, 2010).

A growing line of literature in macroeconomics referred to as "rational inattention" utilizes the mutual information for measuring the amount of information retained by economic agents for decision-making (Sims, 2003, 2006, 2011; Woodford, 2009). Given the limited capacity of individuals to process information, economic agents choose optimally the amount of information to retain. In a discussion by Mankiw et al. (2004), it is mentioned that disagreement among the economic forecasters may be explicated in terms of their differences in retaining information. We have not seen a formal justification for this intuition.

From this overview, we can conclude that, adopting Kullback's (1959) words, "there is currently heterogeneous development of statistical procedures" for measuring uncertainty and disagreement of economic forecasters.

3. Information framework

Figure 1 shows the diagram of the information framework for the notions of uncertainty, information, and disagreement. Forecast distributions are generated in an environment, denoted by X. We assume that X is stochastic and endowed by a probability distribution G with a generalized density g relative to a measure $\nu(x)$. This assumption is needed only for the information node in Fig. 1, however, it is applicable to the entire framework. An economic forecaster refers to a forecast distribution which can be the subjective distribution of an individual i solicited through a survey or the predictive distribution of an econometric model. The forecaster i uses an outcome x_i of X and provides the postdata distribution $f_{y|x_i}$. In this case, G can be the probability distribution of X with $g(x_i) = P(X = x_i)$ or the weight $w_i = g(x_i)$ given to the forecaster i. Figure 1 illustrates this case for the ease of exposition, which is not germane for the framework. More generally, economic forecasters can be models where X represents the various subsets of predictors of Y and g is discrete, or X is a random regressor for predicting Y and g is a continuous density.

3.1. Uncertainty

In Fig. 1, the left branch maps each forecast distribution by an uncertainty function $\mathcal{U}(f_{y|x_i})$ to a real number. The uncertainty function measures the lack of concentration of the forecast density.

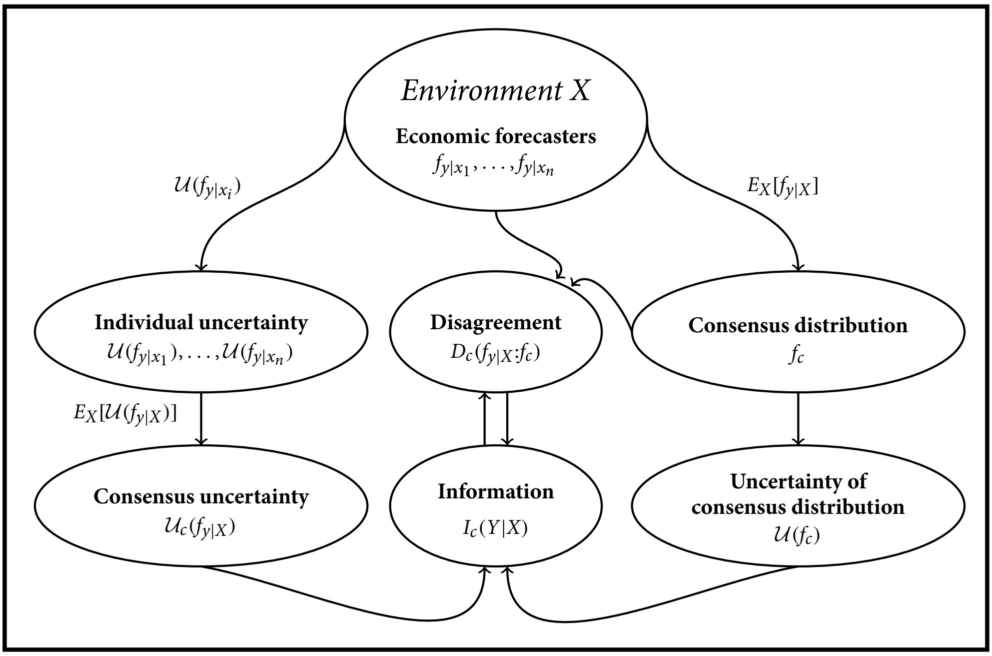

Figure 1. Schematic representation of the information framework for uncertainty and disagreement of economic forecasters.

More concentrated distributions are associated with lower uncertainty and more information about predicting Y. Two desirable properties of an uncertainty function are as follows (Ebrahimi et al., 2010):
(a) $\mathcal{U}(f) \leq \mathcal{U}(f^*)$ where f^* is a proper or improper uniform density; and
(b) $\mathcal{U}(f)$ is a concave function of f.
According to (a), the most uncertain case is when all outcomes of Y seem equally likely, hence no outcome can be favored over the others by a higher probability.

As shown in Fig. 1 (right branch), the individual forecast distributions are pooled into a *consensus forecast distribution* by the mixture model

$$f_c(y) = \int f_{y|x}(y)g(x)d\nu(x), \tag{4}$$

where $d\nu(x) = dx$ for the continuous case and $d\nu(x) = 1$ and the integral becomes summation for the discrete case. The case of $w_i = g(x_i)$ gives the finite mixture (3) and when $G(x)$ is the uniform distribution gives the special case (1), previously noted by Giordani and Söderlind (2003) and Wallis (2005). We use f_c as an estimate of the distribution of Y, but at this point, we avoid introducing additional notations and present the concepts assuming that f_c represents the distribution of Y. In this formulation, the uncertainty about Y is measured by the uncertainty of consensus distribution $\mathcal{U}(f_c)$.

An alternative aggregation strategy, shown in the left branch, is to obtain the *consensus uncertainty* by pooling the uncertainty functions of the forecast distributions

$$\mathcal{U}_c(f_{y|X}) = \int \mathcal{U}(f_{y|X})g(x)d\nu(x). \tag{5}$$

In Fig 1, (4) and (5) are presented as expectations, however, the assumption of stochastic X is not necessary for pooling and $g(x_i) = w_i, i = 1, \ldots, n$ can be weights assigned to the forecasters.

The left and right branches in Fig. 1 are connected by two related measures as described next.

3.2. Information

In Fig. 1, the *information* node, which connects the left and right branches, is defined by the expected uncertainty reduction:

$$I_c(Y|X) = E_X\Big[\mathcal{U}(f_c) - \mathcal{U}(f_{y|X})\Big] = \mathcal{U}(f_c) - \mathcal{U}_c(f_{y|X}) \geq 0. \tag{6}$$

The non-negativity of the expected information is implied by the property of the mixture, where the concave functional $\mathcal{U}(\cdot)$ of the mixture density (4) is greater than the same mixture of the concave functional of the forecast densities (5). The concavity of $\mathcal{U}(\cdot)$ is a reasonable requirement, since on average, one variable (X) reduces the uncertainty about another variable Y, with the worst case being noninformative (DeGroot, 1962).

The inequality in (6) becomes equality, if and only if $f_{y|x}(y) = f_c(y)$ for all (x, y); i.e., X and Y are stochastically independent. For this implication, the assumption of stochastic X is needed. Thus, $I_c(Y|X)$ measures the amount of information that the individual forecasters draw from the environment, which is not captured by the consensus distribution.

3.3. Disagreement

In Fig. 1, the *disagreement* node, which directly connects the forecast distributions to the consensus model, also connects the left and right branches indirectly through the information node. The disagreement is defined in terms of the discrepancy between an individual forecast distribution $f_{y|x}$ and the consensus forecast distribution f_c, measured by an information divergence function $D(f_{y|x} : f_c) \geq 0$. The divergence function is convex in $f_{y|x}$ and the equality holds if and only if $f_{y|x}(y) = f_c(y)$ for all y. The disagreement between forecasters shown in Fig. 1 is the aggregate divergence function given by the expected information divergence:

$$D_c(f_{y|X} : f_c) = \int D(f_{y|X} : f_c) g(x) d\nu(x) \geq 0, \tag{7}$$

where the inequality becomes equality, if and only if $f_{y|x}(y) = f_c(y)$ for all (x, y). In Fig. 1, (7) is presented as an expectation, but the assumption of stochastic X is not necessary for pooling the individual divergence measures $D(f_{y|x_i} : f_c), i = 1, \ldots, n$. With the stochastic assumption, $D_c(f_{y|X} : f_c) = 0$, X and Y are stochastically independent.

Traditionally, disagreement is measured by the spread among the point forecasts. A point forecast can be the mean, median, or mode of the forecast distribution (Engelberg et al., 2009). The equality of point forecasts does not imply the equality of forecast distributions. However, any spread among such point forecasts is captured by $D_c(f_{y|X} : f_c) > 0$ and $D_c(f_{y|X} : f_c) = 0$ implies that all such point forecasts are equal. Even when all the point forecasts are equal, still other features of the forecast distributions (spreads around the point forecasts, shapes) could be different, which will be captured by $D_c(f_{y|X} : f_c) > 0$. Hence, the information disagreement (7) provides *strong measures of disagreement*.

Both (6) and (7) measure the discrepancy between the individual forecasters and the consensus distribution. Therefore, it is reasonable to select D and $\mathcal{U}$ such that $D_c(f_{y|X} : f_c)$ and $I_c(Y|X)$ be increasing functions of each other. In Fig. 1, the two arrows between the disagreement and the information represent the one-to-one relationships between these two notions.

3.4. Jensen–Shannon divergence

Many examples of $\mathcal{U}(f)$ and $D(f_{y|x} : f_c)$ are available which can be used for implementing the information framework. The main uncertainty function in the information theory is the Shannon entropy defined by:

$$\mathcal{U}(f) = H(f) = H(Y) = -\int f_y(y) \log f_y(y) d\nu(y), \tag{8}$$

provided that the integral exists. The mixture model set the stage for implementation of the Jensen–Shannon divergence. The JS of a finite mixture model is defined by:

$$JS(f_c; g) = H(f_c) - \sum_{i=1}^{n} g(x_i)H(f_{y|x_i}) \geq 0. \tag{9}$$

The JS of a continuous mixture is defined similarly. The non-negativity of (9) is implied by concavity of $H(\cdot)$ and the inequality becomes equality if and only if $f_{y|x_i}(y) = f_c(y)$ almost everywhere. The JS admits KL and mutual information representations. It has been used in various applications in several fields, however, until recently it was unnoticed in statistics and econometrics; see Beheshti et al (2016) and references therein.

The JS connects the left and right branches in Fig. 1 as follows:

$$JS(f_c; g) = I_c(X, Y) = M(X, Y) \tag{10}$$

$$= D_c(f_{y|X} : f_c) = K_c(f_{y|X} : f_c), \tag{11}$$

where

$$M(X, Y) = E_X\Big[H(f_c) - H(f_{y|X})\Big] \geq 0, \tag{12}$$

is the *mutual information*,

$$K_c(f_{y|X} : f_c) = E_X[K(f_{y|X} : f_c)] \geq 0, \tag{13}$$

and

$$K(f_{y|x} : f_c) = \int f_{y|x}(y) \log \frac{f_{y|x}(y)}{f_c(y)} d\nu(y) \geq 0, \tag{14}$$

provided that the integral exists, is the KL divergence. For the finite mixture model (3), $f_i(y) = f_{y|x_i}(y)$ is the conditional forecast density with probability $w_i = g(x_i)$.

Representation (13) gives the KL information disagreement between the individual forecast distribution $f_{y|x}$ and the consensus forecaster distribution f_c. This measure is well-defined because $f_c(y) = 0$ for any y implies that $f_{y|x}(y) = 0$ for all x. The inequality in (14) becomes equality if and only if $f_{y|x}(y) = f_c(y)$ for all y. The inequality in (13) becomes equality if and only if $f_{y|x}(y) = f_c(y)$ for all (x, y), hence when Y and X are independent.

Representation (12) gives the expected uncertainty reduction (6) in terms of the mutual information between X and Y:

$$M(X, Y) = H(Y) - H_c(Y|X) \tag{15}$$

$$= H(X) - H_y(X|Y), \tag{16}$$

where $H_c(Y|X) = E_X[H(f_{y|X})]$ is the *conditional entropy*, $H(X)$ is the entropy of the environment variable, and $H_y(X|Y)$ is the conditional entropy of X given Y. The mutual information is a key measure in the Bayesian information theory and in the communication information theory. In the Bayesian information theory, by representation (15), $M(X, Y)$ is the expected utility of the conditional distribution $f_{y|x}$ for prediction of Y (Bernardo, 1979a). In the present context, $M(X, Y)$ is the expected utility of using the individual density forecasts (conditional distributions).

Sims (2003, 2006, 2011) has adopted the communication theory interpretation of the mutual information in his economic theory of rational inattention. In this theory, the mutual information is used to measure the expected information retained by an economic agent from the environment. In the present context, the interest lays on measuring the amount of information about the environment provided by the forecasts. In (16), the information transmission channel is the conditional distribution of the environment X, given the forecast. In this representation, $M(X, Y)$ is the expected amount of information that a forecast provides about the environment. The equality $M(X, Y) = K_c(f_{y|X} : f_c)$

given in (10) and (11) explicates the disagreement among economic forecasters in terms of the theory of rational inattention and provides a formal confirmation of the intuition expressed in the discussion of Mankiw et al. (2004).

The following example illustrates the KL and mutual information representations of the JS measure in a simple case.

Example 1. Consider two forecasters $i = 1, 2$ who provide opposite distributions on the binary support for the inflation rate to go up or down, $m = 2$.

$$\text{Forecaster 1: } f_{y|x_1}(y) = p^y(1-p)^{1-y},\ y = 0, 1,\ 0 \le p \le 1;$$
$$\text{Forecaster 2: } f_{y|x_2}(y) = p^{1-y}(1-p)^y,\ y = 0, 1,\ 0 \le p \le 1.$$

For $g_i = P(X = x_i) = .5, i = 1, 2$, the consensus forecast distribution is uniform $f_c(y) = .5, y = 0, 1$. Noting that $H(f_{y|x_1}) = H(f_{y|x_2}) = H_c(f_{y|X})$ and $H(f_c) = \log 2$, straightforward calculations give

$$K_c(f_{y|X} : f_c) = M(X, Y) = \log 2 + p \log p + (1-p) \log(1-p).$$

Figure 2 illustrates the relationship between disagreement and information provided by these two forecast distributions as functions of p. When p is close to 0.5, say $0.4 < p < 0.6$, there is a little disagreement between the two forecasters and the consensus distribution f_c represents the individuals very well. In these situations, the vertical distance between $H(f_c)$ and $H_c(f_{y|X})$ is small and the $f_{y|x_i}$'s do not provide much information over and above the consensus distribution, hence $K_c(f_{y|X}, f_c)$ is negligible. As p departs from 0.5, the disagreement between the two forecasters increases and the individual forecast distributions become more concentrated at the opposite directions and become more informative about the binary variable, hence $K_c(f_{y|X}, f_c)$ becomes substantial.

The information framework may also be implemented using generalizations of Shannon entropy and KL divergence. However, caution should be exercised because: (a) not all generalizations of Shannon entropy are concave, so can produce negative values for (6); and (b) generalizations of the KL divergence measures do not admit representations in terms of uncertainty reduction, so can produce different values for (6) and (7). Generalized entropy and divergence measures can be used only when (6) and (7) are monotone increasing functions of each other. The following example illustrates some of these issues.

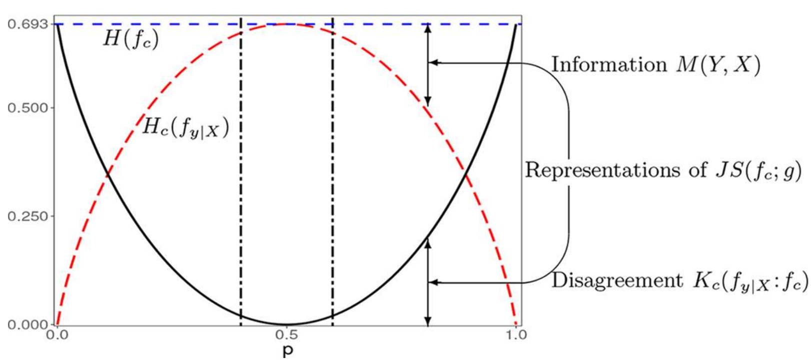

Figure 2. The information provided by disagreement between two Bernoulli forecast distributions with parameters p and $1 - p$.

Example 2. Consider the case of random regressor $f_{y|x} = N(\mu_{y|x}, \sigma^2_{y|x})$, where $\mu_{y|x} = \alpha + \beta x$, and X is continuous with $g(x) = N(\mu_x, \sigma^2_x)$. Then the joint distribution is the bivariate normal with correlation parameter ρ and the "consensus" forecast model is the marginal distribution $f_y(y) = N(\mu_y, \sigma^2_y)$.

(a) The mutual information for any bivariate normal distribution with correlation parameter ρ is

$$K_c(f_{y|X} : f_y) = M(X, Y) = -0.5 \log(1 - \rho^2).$$

(b) Rényi information divergence (Rényi, 1961) is defined as

$$K_r(f_{y|x} : f_c) = \frac{1}{r-1} \log \int f^r_{y|x}(y)[f_c(y)]^{1-r} d\nu(y), \quad r \neq 1, \quad r > 0.$$

For $r = 1, K_r(f_{y|x} : f_c) = K(f_{y|x} : f_c)$. The Rényi entropy $H_r(f)$ is given by $-K_r(f : 1)$ and $H_1(f) = H(f)$. Hirschberg et al. (1991) and Granger et al. (2004) have argued favorably about $K_{1/2}(\cdot, \cdot)$ because it is symmetric in its arguments and is related to Hellinger and Bhattacharya distances and the Matusita index of affinity. The Rényi measures for (6) and (7) are as follows:

$$E_X[H_r(Y) - H_r(Y|X)] = M(X, Y) = -\frac{1}{2} \log(1 - \rho^2), \tag{17}$$

$$E_X[K_r(f_{y|X} : f_y)] = M(X, Y) - \frac{\log(1 + (r-1)\rho^2)}{2(r-1)} + \frac{r\rho^2}{2(1 + (r-1)\rho^2)} \tag{18}$$

(details are given in the Appendix). Thus, Rényi entropy divergence provide different measures for the uncertainty reduction (6) and disagreement (7) which are monotonically related. The uncertainty reduction (17) is not a function of r, while the Rényi divergence is continuous and increasing in r (van Erven and Harremoös, 2014). The measures in (17) and (18) are also increasing functions of ρ^2. Hence, as ρ^2 increases, the disagreement increases and the forecast models become more informative than the consensus model (marginal distribution of Y). This can be seen by noting that the conditional distributions become more concentrated around their means that vary along the regression line. Also, Rényi entropy is concave for $r < 1$ (van Erven and Harremoös, 2014), thus unlike the case of the normal model, in general, the non-negativity of information measure (6) in terms of H_r is not guaranteed when $r > 1$; see Ebrahimi et al. (2010) for an example.

4. Dynamics of inflation uncertainty

In the SPF database, at quarter t, forecast distributions for inflation are solicited for a set of m_t preassigned intervals in the form of probability vectors

$$f_{y_{t+h}|x_{it}} = \mathbf{p}_{i(t+h)} = \Big(p_1(y_{t+h}|x_{it}), \ldots, p_{m_t}(y_{t+h}|x_{it})\Big), \quad i = 1, \ldots, n, \quad h = .5, 1.5, \ldots, 7.5.$$

The data are collected in the middle of each quarter for the inflation rates of current and following years. The forecast horizons of $h = 0.5 - 3.5$ are for the current year and $h = 4.5 - 7.5$ are for the following year. The inflation rate is measured by the GDP deflator between the consecutive years.

As in the case of computing variance, various methods can be used for computing the entropy based on the grouped data; e.g., the entropy of the probability vector $H(\mathbf{p})$ provides a crude entropy measure for the grouped data, analogous to the variance of the mid-intervals. Other alternatives include the entropy of best density fit to the histogram (Hall, 1990), the entropy of the histogram density proposed by Hall and Morton (1993) as an entropy estimate for the continuous case, the Bayes estimate of entropy based on the histogram partition (Mazzuchi et al., 2008), maximum entropy based on the moments, and expected entropy under Dirichlet embedding. In this section, we use $H(\mathbf{p})$. The maximum entropy and Dirichlet embedding will be presented in Sections 5 and 6.

4.1. Adjustment for the number of intervals

For studying the SPF data over time, consideration should be given to several changes made since 1968. Table 1 displays the intervals (bins) for the SPF database. The number of bins, interval widths, and the overall range have been changed a few times. Within the periods 1968Q4-1981Q2 and 1981Q3-1991Q4, the changes are location shifts. The ranges and the number of interval remain unchanged, hence measures of uncertainty within these periods are comparable. Rich and Tracy (2010) combined pairs of one percent intervals of 1968–1981 and post 1992 into intervals of width 2% to match the 1981–1991 period. The width of the intervals does not effect the entropy of the probability vector, but the number of intervals does affect the entropy. Adjustment by reducing the number of intervals reduces entropy, while leaving the important issue of the unequal number of intervals unresolved.

The information theoretic method of adjustment for the number of intervals is the normalized entropy index for discrete variable $H(\mathbf{p}_{i(t+h)})/\log m_t$, where $\log m_t$ is the entropy of uniform distribution on m_t bins during the period t (Golan et al., 2000). The discrete entropy is not sensitive to the range of the support, but the continuous entropy is. However, the adjustment for the number of bins is not applicable to the continuous case.

Figure 3 shows the time series patterns of the consensus uncertainty $H_c(\mathbf{p}_{i(t+h)})$ and the uncertainty of the consensus distribution $H(\mathbf{p}_{c(t+h)})$ before and after adjustments for the changes made in the number of bins in 1981Q3 and 1992Q1 (shown by the vertical lines). Following the literature, we have used $g_i =$

Table 1. Bins for SPF distributions.

Period t	1968Q4-1973Q1	1973Q2-1974Q3	1974Q4-1981Q2	1981Q3-1985Q1	1985Q2-1991Q4	1992Q1-Present
Bins	$\geq$ 10%	$\geq$ 12%	$\geq$ 16%	$\leq$ 12%	$\leq$ 10%	$\leq$ 8%
	9–9.9%	11–11.9%	15–15.9%	10–11.9%	8–9.9%	7–7.9%
	8–8.9%	10–10.9%	14–14.9%	8–9.9%	6–7.9%	6–6.9%
	⋮	⋮	⋮	⋮	⋮	⋮
	$\leq$ −3%	$\leq$ −1%	$\leq$ 3%	$\leq$ 4%	$\leq$ 2%	$\leq$ 0%
Number of bins m_t	15	15	15	6	6	10

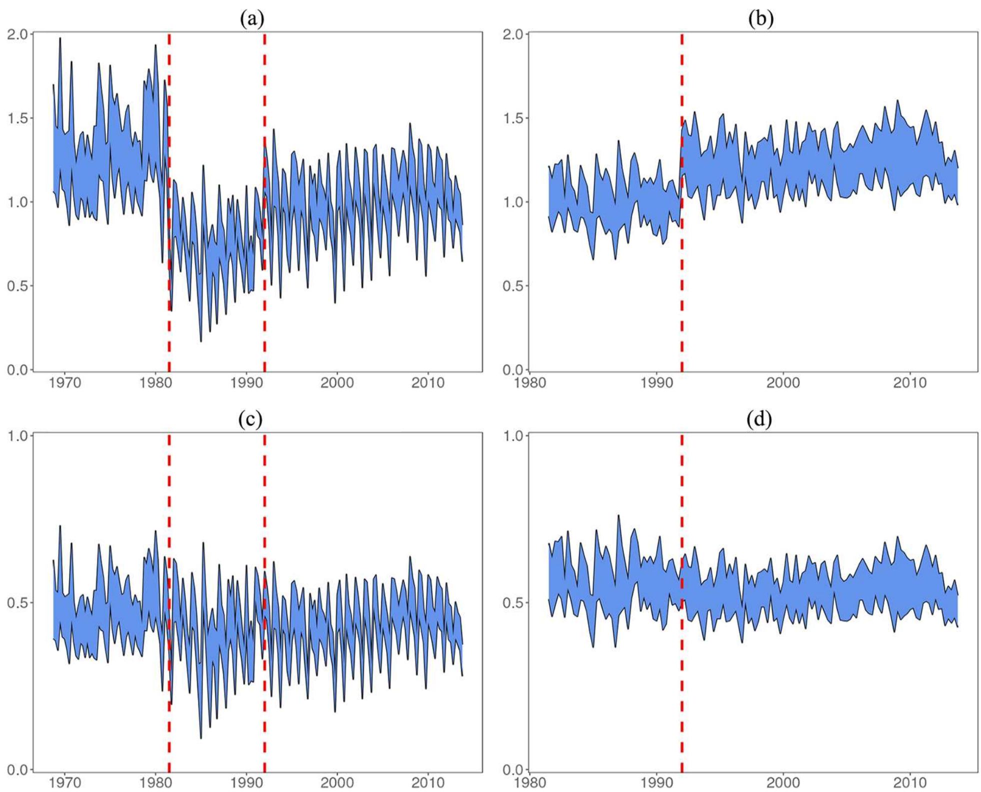

Figure 3. Time series plots of uncertainty and disagreement (mutual information) about the current and following years' inflation rates; vertical lines indicate dates that the number of bins are changed (the series in the lower panels are adjusted for the number of bins). (a) Measures for the current year (unadjusted), (b) measures for the following year (unadjusted), (c) measures for the current year (adjusted), and (d) measures for the following year (adjusted).
Notes: Upper series, the entropy of the consensus forecast distribution; lower series, the consensus entropy of the individual distributions; colored area disagreement (mutual information).

$1/n, i = 1, \ldots, n$ for aggregation. The plots are for the current year's inflation rate (left panels) and the following year's inflation rate (right panels). The plots for the unadjusted measures (upper panels) show shift when the number of bins changes and the adjustments for the number of bins by the entropy index (lower panels) correct for the shifts. Both panels show seasonal patterns of $H_c(\mathbf{p}_{i(t+h)})$ and $H(\mathbf{p}_{c(t+h)})$. The difference in forecast horizons of the survey in each quarter leads to these seasonal patterns. We also note that the quarterly aggregate uncertainty about the current year is generally lower than the aggregate uncertainty about the following year.

In each panel, the difference between the two series depicts the information provided by the forecast environment X about the inflation rate Y, which also measures the disagreement. Figure 4 shows unadjusted and adjusted plots of the information disagreement. The unadjusted plot for the current year shows higher levels of disagreement for the early period, but when adjusted for the number of bins, the level of disagreement stabilizes. The plots for the current year's and following year's forecasts indicate little difference of disagreement among the forecasters over time.

4.2. Models for inflation uncertainty

Relationships between the uncertainty with disagreement and perceived inflation have been examined using various measures and models which have led to contradictory findings (Zarnowitz and Lambros,

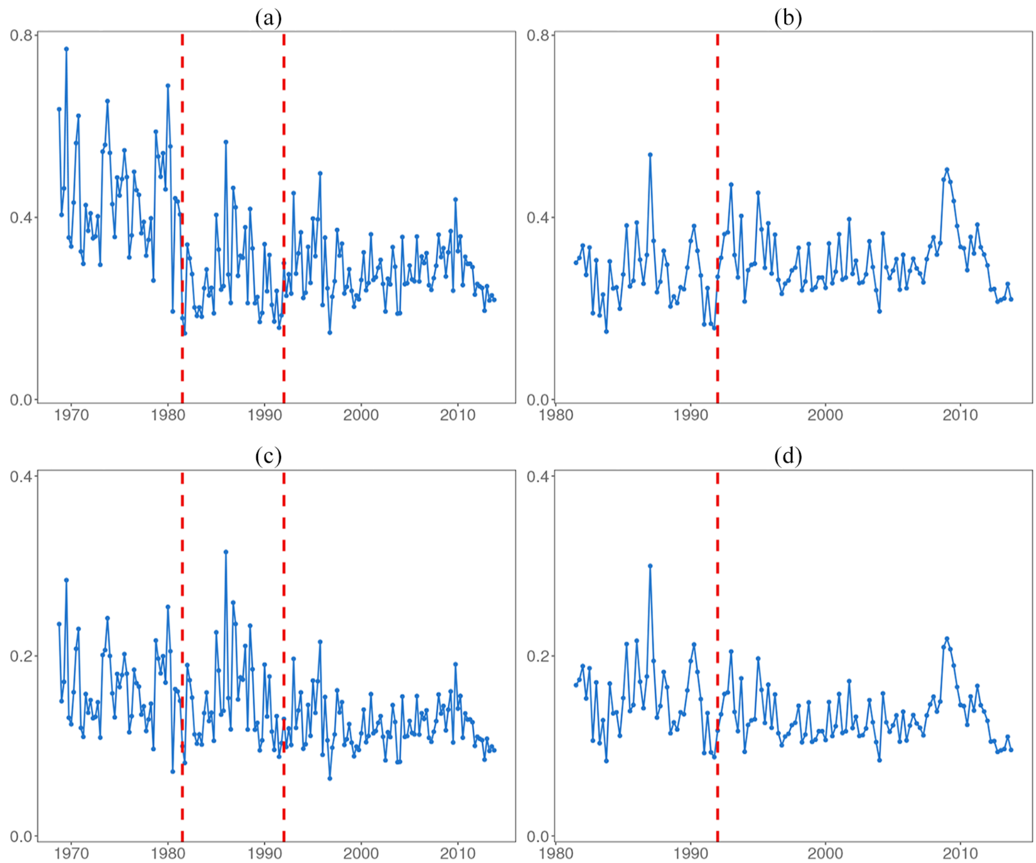

Figure 4. Time series plots of disagreement (mutual information) about the current and following years' inflation rates; vertical lines indicate dates that the number of bins are changed (the series in the lower panels are adjusted for the number of bins). (a) Disagreement current year (unadjusted), (b) disagreement following year (unadjusted), (c) disagreement current year (adjusted), and (d) disagreement following year (adjusted).

1987; Giordani and Söderlind, 2003; Lahiri and Sheng, 2010; Rich and Tracy, 2010). Some authors have asserted that the dispersion of point forecasts is driven by the uncertainty about the economic variable, and hence can be used for measuring the uncertainty (Barnea et al., 1979; Levi and Makin, 1980; Bomberger and Frazer, 1981; Bomberger, 1996; Makin, 1982; Giordani and Söderlind, 2003). An alternative view asserts that the dispersion of point forecasts is not necessarily reflective of the variance of their perceived distributions of the economic variable (Zarnowitz and Lambros, 1987; Lahiri et al., 1988). Okun (1971) as well as Friedman's (1977) Nobel Lecture both considered the interrelationship between inflation uncertainty and perceived inflation.

We examine the association between the uncertainty and dispersion of point forecasts to shed light on the contradictory results reported in the literature (Zarnowitz and Lambros, 1987; Giordani and Söderlind, 2003). We also test the association between the inflation rate uncertainty and the anticipated inflation hypothesized by Okun (1971) and Friedman (1977), beyond the dynamic of the uncertainty about the inflation rate.

Thus far, the empirical models developed based on the SPF data are for the quarterly aggregate level uncertainty and do not include the dynamics of the quarter-to-quarter evolution of the uncertainty. Table 2 shows the autocorrelation of $H_c(\mathbf{p}_{i(t+h)})$ and $H(\mathbf{p}_{c(t+h)})$ after the seasonality is removed. These observations suggest that the forecast uncertainty about the inflation is not memoryless. We develop Bayesian hierarchical models for the uncertainty of the inflation rate for the individual forecaster and aggregate levels. Our models include the dynamics of the uncertainty about the inflation.

We use the SPF survey for 1992Q1-2013Q4, because after 1992Q1, the SPF provides the "cleanest data" (Engelberg et al., 2009). During this period, the number of categories remains constant, thus adjustment for the number of categories is not needed. The forecast distributions before 1981Q3 usually are for the percent changes for the current year, but in some periods are for the percent change in the following year. There are some suspicions that the data for the first quarters of 1985 and 1986 do not follow the standard forecast distributions of the current year and the following year. Before 1992Q1, the forecast distributions are based on the percentage change in GNP price deflator, but after 1992Q1, the base was switched to the percent change in the price index for GDP. Several papers have used the data since 1968 deleting the problematic quarters and Engelberg et al. (2009) used the data after 1991.

4.2.1. Individual level model

We consider the following model for the uncertainty about the inflation rate:

$$U_{it} = \alpha + \rho U_{i(t-1)} + \sum_{k=1}^{3} \beta_k Q_{kt} + \theta \mu_{it} + \epsilon_{it}, \tag{19}$$

where $U_{it} = H(\mathbf{p}_{i(t+h)})$ is the measure of inflation uncertainty and μ_{it} is the inflation point forecast of the individual i at time t, Q_{kt} is the indicator variable for the quarter k, ϵ_{it}'s are conditionally independent $\epsilon_{it}|\mathcal{Z}, \sigma^2 \sim N(0, \sigma^2)$; here $\mathcal{Z}$ denotes the set of all regressors and parameters.

The first three terms in (19) capture the dynamic of the inflation rate uncertainty over time. The uncertainty in the preceding quarter $U_{i(t-1)}$ is individual-specific and captures the forecasting pattern of the forecaster. The quarter indicators capture the forecast horizon and quarterly variation of the

Table 2. Autocorrelation of uncertainty about inflation rates after the seasonality is removed.

	$t-1$	$t-2$	$t-3$	$t-4$	$t-5$
Current year forecast distribution					
Consensus uncertainty $H_c(\mathbf{p}_{i(t+h)})$	0.39	0.22	0.28	0.26	0.27
Uncertainty of consensus distribution $H(\mathbf{p}_{c(t+h)})$	0.37	0.19	0.16	0.15	0.17
Following year forecast distribution					
Consensus uncertainty $H_c(\mathbf{p}_{i(t+h)})$	0.20	0.30	0.07	−0.10	0.02
Uncertainty of consensus distribution $H(\mathbf{p}_{c(t+h)})$	0.43	0.41	0.25	0.07	0.08

knowledge about the inflation rate. The summation can be interpreted as the "aggregate shock" *á la* Lahiri and Sheng (2010). The term μ_{it} is for testing the relationships between the uncertainty about the inflation rate and the anticipated inflation.

We use the following Bayesian hierarchical specification:

$$\begin{aligned}&\alpha|\mu_0,\tau \sim N(\mu_0,\tau^2),\ \ \mu_0 \sim \text{Uniform}[0,\ \log m],\ \ \tfrac{1}{\tau^2} \sim \text{Gamma}(0.001, 0.001),\\ &\rho \sim \text{Uniform}(-1,\ 1),\ \ \beta_k \sim N(0, 1000),\ \ k = 1, 2, 3,\ \ \theta \sim N(0, 1000),\\ &\tfrac{1}{\sigma^2} \sim \text{Gamma}(0.001, 0.001).\end{aligned} \tag{20}$$

For estimation, we have used the data of individual forecasters with at least one block of nine consecutive quarters of forecast distributions since 1992Q1. The sample for the current year's forecast includes $n = 68$ individuals and 1,599 observations and for the following current year's forecast includes $n = 65$ individuals and 1,526 observations. With these sample sizes, the effects of the priors are negligible. Computations are implemented using WinBUGS.

Table 3 summarizes the posterior results for the respondent's uncertainty about the current and following year inflation rate. The results indicate that the dynamics of uncertainty about the current year and following year inflation rates are essentially the same. The 95% intervals for the autoregressive coefficient indicate that in each quarter, the uncertainty about the inflation rate is positively associated with the previous quarter's uncertainty. The 95% intervals for the coefficients of the seasonal factors show seasonal variations of uncertainty about the current year's inflation rate. The decreasing magnitudes of the seasonal effects reveal that respondent's uncertainty about the current year's and the following year's inflation rates decrease during the year. This confirms that the respondents have more information about the economy in the later quarters than the earlier quarters of a year. The 95% intervals for the coefficient of the expected forecast shows that higher perceived inflation by respondents accompanied with higher uncertainty about inflation. This confirms Friedman's (1977) view who argues that higher rate of inflation and anticipated inflation could lead to higher inflation uncertainty.

4.2.2. Aggregate-level model

For the aggregate uncertainty about the inflation, we include additional terms in (19) as follows:

$$U_t = \alpha + \sum_{j=1}^{J} \rho_j U_{t-j} + \sum_{k=1}^{3} \beta_k Q_{kt} + \theta_1 \bar{\mu}_t + \theta_2 \hat{H}(\mu_t) + \epsilon_t, \tag{21}$$

where U_t denotes an aggregate measure of uncertainty ($H_c(\mathbf{p}_{i(t+h)})$ or $H(\mathbf{p}_{c(t+h)})$) at time t, α and Q_{kt} are defined as above, $\bar{\mu}_t = \frac{1}{n}\sum_{i=1}^{n} \mu_{it}$ is the consensus forecast (average of the point forecasts), $\hat{H}(\mu_t) = 0.5 + 0.5\log(2\pi) + 0.5\log\sigma^2_{\mu_t}$ is the entropy of the maximum entropy normal model for the underlying distribution of point forecasts $\mu_t \sim N(\bar{\mu}_t, \sigma^2_{\mu_t})$ (in effect, it is the same as the log of the variance of point forecasts traditionally used), and ϵ_ts are conditionally independent $\epsilon_t|\mathcal{Z}, \sigma^2 \sim N(0, \sigma^2)$; here $\mathcal{Z}$ denotes the set of all regressors and parameters.

Table 3. Posterior results for the individual forecaster uncertainty about the inflation rate.

	Current year inflation*		Following year inflation**	
	Mean	95% interval	Mean	95% interval
α	−0.01	[−0.06, 0.05]	0.11	[0.06, 0.16]
ρ	0.65	[0.61, 0.69]	0.77	[0.74, 0.81]
β_1	0.47	[0.43, 0.51]	0.15	[0.12, 0.18]
β_2	0.16	[0.12, 0.20]	0.03	[−0.01, 0 .06]
β_3	0.14	[0.10, 0.18]	0.03	[−0.01, 0.06]
θ	0.05	[0.03, 0.07]	0.03	[0.02, 0.05]

*Forecast horizon 0.5–3.5 **Forecast horizon 4.5–7.5

Table 4. Posterior results for the uncertainty about the current year's inflation (forecast horizon 0.5–3.5).

	Consensus uncertainty				Uncertainty of consensus distribution			
	Mean	95% interval	Mean	95% interval	Mean	95% interval	Mean	95% interval
α	0.27	[0.07, 0.48]	.38	[0.16, 0.60]	0.43	[0.18, 0.68]	0.99	[0.79, 1.20]
ρ_1	0.39	[0.18, 0.60]	34	[0.13, 0.55]	0.37	[0.18, 0.57]	0.18	[0.03, 0.33]
β_1	0.48	[0.42, 0.55]	0.46	[0.38, 0.53]	0.48	[0.41, 0.56]	0.35	[0.30, 0.41]
β_2	0.27	[0.21, 0.33]	0.27	[0.21, 0.32]	0.27	[0.20 , 0.34]	0.24	[0.19, 0.28]
β_3	0.18	[0.13, 0.24]	0.18	[0.13, 0.23]	0.17	[0.11, 0.24]	0.16	[0.12, 0.20]
θ_1	0.01	[−0.02, 0.04]			0.02	[−0.02, 0.06]		
θ_2			0.03	[−0.01, 0.06]			0.14	[0.11, 0.17]
DIC	−223.2		−230.0		−189.2		−243.4	

DIC, deviance information criterion

The summations in (21) capture the dynamic of the aggregate uncertainty about the inflation rate over time. The average of the individual point forecasts at time t is used for testing the relationship between the anticipated inflation and inflation uncertainty. The point forecast entropy is included for testing the association between the uncertainty about the inflation rate and the dispersion of the point forecasts at time t.

The prior distributions for the parameters of (21) are the same as (20) with ρ_j's being independently distributed as ρ and $\theta_h, h = 1, 2$ being independently distributed as θ.

Table 4 summarizes the posterior results for the uncertainty about the current year inflation for the consensus uncertainty $U_t = H_c(\mathbf{p}_{i(t+h)})$ and the uncertainty of the consensus distribution $U_t = H(\mathbf{p}_{c(t+h)})$ with $J = 1$. The results for the dynamics of uncertainty are essentially the same for the two measures. The 95% intervals for the autoregressive coefficient indicate that in each quarter, the uncertainty about the inflation rate is positively associated with the previous quarter's uncertainty. The 95% intervals for the coefficients of the seasonal indicators show seasonal variations of uncertainty about the current years' inflation rate. The decreasing magnitudes of the seasonal effects reveal that the level of aggregate uncertainty about the current year's inflation rate decreases during the year. This result reflects the fact that the respondents have more information about the economy in the later quarters than the earlier quarters of a year. (We also considered models with $J = 2, 3, 4$, models without the seasonal indicators, and models that included only the dynamics of the uncertainty. All provided essentially similar results, hence are not reported. The models reported are selected based on the deviance information criterion (DIC); Spigelhalter et al. (2002).)

The 95% intervals for the coefficients of the consensus forecast (θ_1) for both measures of aggregate uncertainty are centered near zero. (We also examined models without the lag term and found the same result). Therefore, unlike the individual level model, at the aggregate level, the data do not provide evidence for the relationship between the anticipated inflation and the inflation uncertainty.

When the dispersion of the point forecasts is included, the results are different for the two measures of aggregate uncertainty. For the consensus uncertainty, the 95% interval for θ_2 includes zero. For the uncertainty of the consensus distribution, the posterior interval for θ_2 indicates a positive association. The DIC indicates substantial improvement of the model fit. (We also calculated approximate Bayes factor for the inclusion of this term which provided decisive evidence). The inclusion of dispersion of the point forecasts has increased the intercept and decreased the effect of Q_1 and the autoregressive coefficient.

5. Moment-based measures

Traditionally, the variance is used for measuring uncertainty. The variance is a concave function, but it is not always bounded above by the variance of the uniform distribution on the support of F (Ebrahimi et al., 2010).

Within the information framework (Fig. 1), the role of variance for measuring uncertainty is best understood in terms of the maximum entropy principle which characterizes the normal distribution by variance. The continuous ME model based on the first two moments of the forecast distributions is normal $f^*_{y|x_i} = N(\mu, \sigma^2)$ and its entropy is given by:

$$H(f^*_{y|x_i}) = \frac{1}{2} + \frac{1}{2}\log(2\pi) + \frac{1}{2}\log\sigma^2_{y|x_i}. \tag{22}$$

Wallis (2006) suggested that the expression at the right-hand side of (22) may be used as an adjustment for "comparing entropy-based and moment-based uncertainty measures" where the "adjustment being subject to the acceptability of the normality assumption." Mitchell and Wallis (2010) interpreted (22) as "the expected logarithmic score of the correct conditional density."

In this case, the marginal distribution of Y is a mixture of normal distributions. For a finite mixture, $f_c(y) = \sum_{i=1}^n g(x_i) f^*_{y|x_i}(y|x_i)$, (12) and (14) do not have closed forms. For implementation of the information framework (Fig. 1), we aggregate the first two moments of the forecast distributions and use the ME normal model with the aggregate variance; e.g, let $g(x_i) = \frac{1}{n}, i = 1, \ldots, n$. Then the aggregate moments are the moments of the consensus distribution (1) and the consensus ME model is normal $f^*_c = N(\mu_c, \sigma^2_c)$. The expected maximum entropy reduction is given by:

$$\Delta H^* = H(f^*_c) - H_c(f^*_{y|X}) = -\frac{1}{2n}\sum_{i=1}^n \log\frac{\sigma^2_{y|x_i}}{\sigma^2_c} \geq 0, \tag{23}$$

where $H_c(f^*_{y|X}) = \frac{1}{n}\sum_{i=1}^n H(f^*_{y|x_i})$. Note that the inequality in (23) cannot be concluded from the concavity of the entropy because $f^*_c = N(\mu_c, \sigma^2_c) \neq \frac{1}{n}\sum_{i=1}^n f^*_{y|x_i}$. However, the inequality in (23) can be seen from (2) by noting that $\sigma^2_{y|x_i} = \sigma^2_i$. In the normal regression terminology, σ^2_c and $\sigma^2_{y|x_i}$ in (23) correspond to the variance of the response variable and error term, respectively.

Furthermore, since f^*_c is not the marginal distribution of Y, (23) is not a mutual information and (12) cannot be concluded. However, an equality analogous to (13) and (15) holds. The well-known KL divergence between two normal distributions gives:

$$\begin{aligned} K^*_c = K_c(f^*_{y|X} : f^*_c) &= \frac{1}{n}\sum_{i=1}^n K(f^*_{y|x_i} : f^*_c) \\ &= \frac{1}{2n}\sum_{i=1}^n \left(\frac{\mu_{y|x_i} - \mu_c}{\sigma_c}\right)^2 + \frac{1}{2n}\sum_{i=1}^n \left[\frac{\sigma^2_{y|x_i}}{\sigma^2_c} - \log\frac{\sigma^2_{y|x_i}}{\sigma^2_c} - 1\right]. \end{aligned} \tag{24}$$

Applying the decomposition (2) of σ^2_c in (24) gives representation (23):

$$K_c(f^*_{y|X} : f^*_c) = H(f^*_c) - H_c(f^*_{y|X}). \tag{25}$$

Representation (24) illustrates $K_c(f^*_{y|X} : f^*_c)$ as a stronger measure of disagreement than the variance of point forecasts. The first term in (24) quantifies the disagreement between the means (point forecasts under the quadratic loss) and the second term quantifies the disagreement between the variances of the forecast distributions. When the means of forecast distributions are all equal (complete agreement on the point forecast), the first term in (24) is zero and the second term measures the disagreement about the spread of their distributions around their common forecast. When the forecasters disagree on the forecast, but with complete agreement on the accuracy of their forecasts, the first term in (24) measures their disagreement and the second term is zero.

The following example illustrates the variance-based measures and compares them with the distribution-based measures for the SPF forecasters.

Example 3. We compute the first two moments of the SPF forecast distributions using the mid-points of intervals and the same width for the intervals at the ends. For the categories that the probabilities were zero, we imputed $p_{ik} = 10^{-4}$ which has negligible effects as compared with the histogram moments.

a) Individual-level variance-based entropies

We use the 2013Q1 and 2013Q4 forecast distributions for the 2013 and 2014 inflation rates to illustrate the moment-based measures for the individual forecasters. The variance-based measures are highly correlated with the entropy of the forecast distribution (the Spearman's rank correlations are all above 0.99). Table 5 shows more results with rows arranged in terms of the forecast horizon. The next three columns show three distribution-based measures: $H(\mathbf{p}_c) = H(\mathbf{p}_{c(t+h)}), H_c(\mathbf{p}_{y|X}) = H_c(\mathbf{p}_{i(t+h)})$, and $JS(f_c; 1/n) = H(\mathbf{p}_{c(t+h)}) - H_c(\mathbf{p}_{i(t+h)})$ which gives KL measure of disagreement and the mutual information. These measures are calculated using the probability vectors given in SPF. The next three columns show three variance-based measures: $H(f_c^*), H_c(f_{y|X}^*)$ and $K_c^* = \Delta H^*$. These measures are calculated using the individual and aggregate variances based on the mid-intervals. The disagreement column illustrates equation (23) which equals to (24), hence (25). These measures formally confirm the common belief that uncertainty increases with the forecast horizon. The last column shows the percent of forecasters ranked equally by the entropy of the forecast distribution $H(\mathbf{p}_{i(t+h)})$ and the variance-based measure, $H(f^*_{y_{t+h}|x_{it}})$. It can be seen that the percentage of forecasters ranked equally decreases with the forecast horizon. Noting that the $H(f^*_{y_{t+h}|x_{it}})$ ranks forecasters according to the variance, we find that as the uncertainty increases along with the forecast horizon, the entropy and variance of the forecast distributions rank forecasters more differently. These results are revealing along the lines of Ebrahimi et al. (1999) that the entropy may be related to high-order moments and offers a much closer characterization of the uncertainty of a distribution than the variance.

b) Aggregate-level variance-based entropies

The upper panels of Fig. 5 show the time series patterns of the variance-based measures (the ME normal model). The patterns for the ME normal distribution are very different from those for the forecast distributions in the upper left panel of Fig. 3. The uncertainty of the consensus distribution (upper boarder) is relatively stable, but the consensus uncertainty (lower border) is quite sensitive to the number of bins. For the current year's forecast, in the middle period where the number of bins is low, the normal entropy drops sharply for fourth quarters because several forecast distributions lie on a single bin. The same pattern is seen in the last period, although to a lower extent. In each upper plot, the difference between the two series depicts the measure of disagreement given by (23) and (24). The lower panels of Fig. 5 show the time series plots of this measure for the current year and following year.

Table 5. Comparison of the rankings of SPF 2013Q1 and 2013Q4 forecasters' uncertainty by the variance-based and distribution-based measures.

			Distribution-based measures			Variance-based ME measures			Forecasters
	n	h	$H(\mathbf{p}_c)$	$H_c(\mathbf{p}_{y\|X})$	$JS(f_c; 1/n)$ $K_c = M$	$H(f_c^*)$	$H_c(f_{y\|X}^*)$	Disagree $K_c^* = \Delta H^*$	equally ranked by $H_c(\mathbf{p}_{y\|X})$ & $H_c(f_{y\|X}^*)$
Current year's inflation									
2013Q4 forecasters	40	0.5	0.86	0.64	0.22	0.91	0.08	0.83	85%
2013Q1 forecasters	38	3.5	1.15	0.91	0.24	1.17	0.93	0.24	37%
Following year's inflation									
2013Q4 forecasters	40	4.5	1.20	0.98	0.22	1.24	0.93	0.31	35%
2013Q1 forecasters	38	7.5	1.26	1.04	0.22	1.28	1.07	0.21	5%

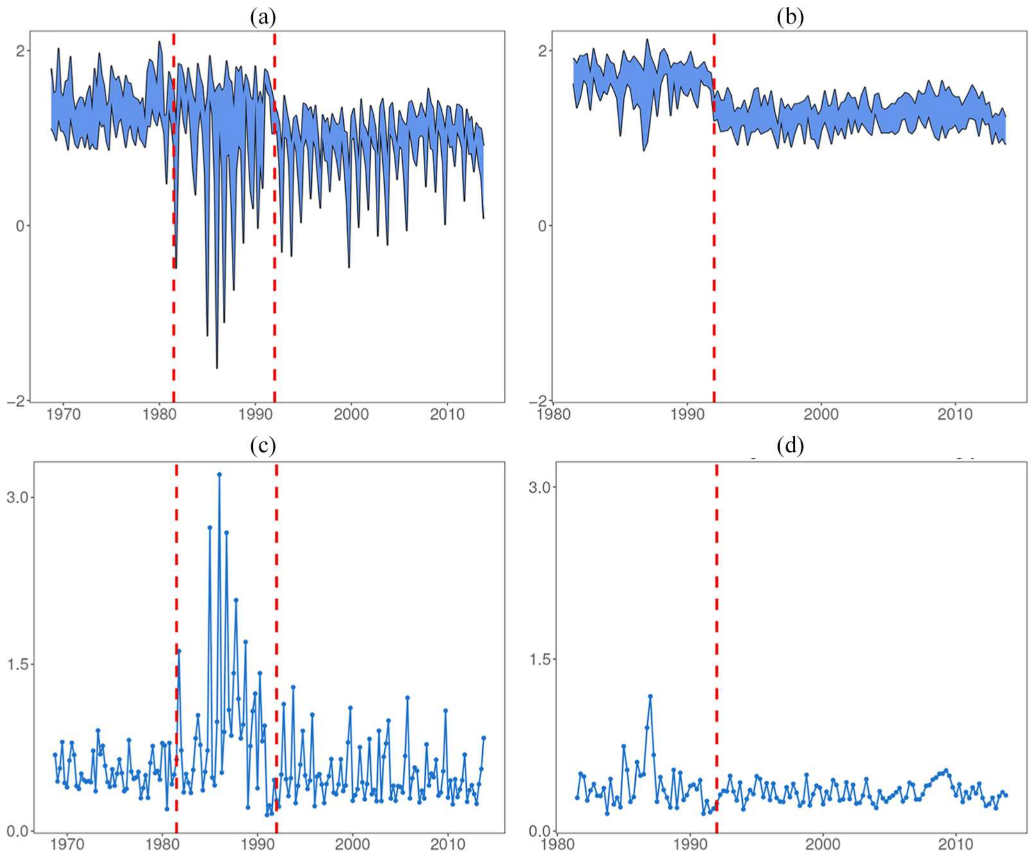

Figure 5. Time series plots of the variance-based measures of uncertainty and disagreement about the current and following year's inflation rates; vertical lines indicate dates that the number of bins are changed. (a) Variance based measures for the current year, (b) variance based measures for the following year, (c) variance based disagreement for the current year, and (d) variance based disagreement for the following year.
Notes: Upper series, the entropy of the consensus forecast distribution; lower series, the consensus entropy of the individual distributions; colored area disagreement (mutual information).

Remark 1. Lahiri et al. (1988) have considered the mean, variance, skewness, and kurtosis of SPF inflation forecast distributions in a regression model for the interest rate. Consideration of characteristics of the forecast distributions beyond the first two moments raises the issue of measuring uncertainty beyond the normal model; e.g., the continuous ME model based on the first four moments is quartic exponential (Zellner and Highfield, 1988). The following proposition gives a general result.

Proposition 1. *Let $f^*_{y|x}$ denotes the individual's ME forecast distribution that satisfies the moment constraints*

$$\mu_{y|x,j} = E_{y|x}(Y^j),\ j = 1, \ldots, J \tag{26}$$

and f^*_a be the ME consensus forecast distribution that satisfies the corresponding aggregate moments

$$\mu_{c,j} = E_g\Big[E_c(Y^j)\Big],\ j = 1, \ldots, J. \tag{27}$$

Then,

$$E_g[K(f^*_{y|X} : f^*_c)] = H(f^*_c) - E_g[H(f^*_{y|X})] \geq 0,$$

where the inequality becomes equality if and only if $\mu_{y|x,j} = \mu_{c,j}$ for all $i = 1, \ldots, n$ and all $j = 1, \ldots, J$.

Proof is shown in the Appendix.

Representations (13) and (15) are analogous to the equality in Proposition 1. However, the ME model f_a^* in Proposition 1 is obtained through the mixture of moments and is not the marginal distribution of $f(y,x)$, hence the average entropy difference in Proposition 1 does not give representations of the mutual information.

6. Uncertainty about the probabilities

A survey solicits estimates of the probabilities

$$\pi_k = P(\xi_{k-1} \leq Y \leq \xi_k) = \int_{\xi_{k-1}}^{\xi_k} f(y)dy, \quad k = 1, \ldots, m,$$

where $\xi_0 = -\infty < \xi_1 < \cdots < \xi_m = \infty$ define a partition of the support of f. The distributions provided by the forecasters $\mathbf{p}_{y|x_i}, i = 1, \ldots, n$ are imprecise and noisy (Boero et al., 2014). Consequently, measures such as the mean, variance, and entropy that are computed as functions of $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_m)$ are also unknown and computed based on $\mathbf{p}_{y|x_i}$ are imprecise and noisy. Boero et al. (2014) refers to this phenomenon as "uncertain uncertainty."

We construct probability distributions for the unknown probability vector $\boldsymbol{\pi}$ based on distributions provided by the forecasters $\mathbf{p}_{y|x_i}, i = 1, \ldots, n$. For each economic forecaster, we have a Dirichlet distribution for $\boldsymbol{\pi}$ with probability density function

$$\phi(\boldsymbol{\pi}|\boldsymbol{\alpha}_i) = \frac{\Gamma(B)}{\Gamma(\alpha_{i1}) \cdots \Gamma(\alpha_{im})} \pi_1^{\alpha_{i1}-1} \cdots \pi_m^{\alpha_{im}-1}, \quad \pi_k \geq 0, \quad \sum_{k=1}^{m} \pi_k = 1,$$

where $\boldsymbol{\alpha}_i = (\alpha_{i1}, \ldots, \alpha_{im})$ is the vector of shape parameters, and $B = \sum_{k=1}^{m} \alpha_{ik}$ is the *degree of belief* parameter. In the Dirichlet embedding by Soofi et al. (2009), the forecaster's probability $\mathbf{p}_{y|x_i}$ represents a feature of $\phi_i(\boldsymbol{\pi})$ such as the mean vector $\hat{\boldsymbol{\pi}} = E(\boldsymbol{\pi}|\mathbf{p}_{y|x_i}) = \mathbf{p}_{y|x_i}$, or the mode vector $\tilde{\boldsymbol{\pi}}_i = \mathbf{p}_{y|x_i}$. Setting the probabilities provided by each economic forecaster $\mathbf{p}_{y|x_i}$ as the mean or mode of the Dirichlet distribution gives the parameters of $\phi(\boldsymbol{\pi}|\mathbf{p}_{y|x_i})$ respectively, as follows:

$$\hat{\boldsymbol{\alpha}}_i = B\mathbf{p}_{y|x_i}, \quad i = 1, \ldots, n, \tag{28}$$

$$\tilde{\boldsymbol{\alpha}}_i = (B - m)\mathbf{p}_{y|x_i} + 1, \quad i = 1, \ldots, n. \tag{29}$$

In this context, B signifies the degree of belief in the accuracy of the probabilities provided by the economic forecaster. The variance of π_k is

$$\text{Var}(\pi_k) = \frac{\alpha_{ik}(B - \alpha_{ik})}{B^2(B+1)}, \quad k = 1, \ldots, m.$$

As B increases, the variances decrease. Thus, in both cases (28) and (29), as B increases the Dirichlet distribution $\phi(\boldsymbol{\pi}|\mathbf{p}_{y|x_i})$ becomes more concentrated at $\mathbf{p}_{y|x_i}$; the ultimate certainty case of $B = \infty$ reflects no uncertainty about $\mathbf{p}_{y|x_i}$, hence taking the forecaster's probabilities at their face values without any uncertainty. In the modal case (29), as $B \to m$ the parameters $\tilde{\alpha}_{ik} \to 1$, $k = 1, \ldots, m$, and Dirichlet distribution $\phi(\boldsymbol{\pi}|\mathbf{p}_{y|x_i})$ becomes nearly uniform, reflecting maximum uncertainty about the forecaster's probabilities, hence no worth for $\mathbf{p}_{y|x_i}$.

In Bayesian terms, one may think of the embedding in terms of the data vector $\mathbf{p}_{y|x_i}$ as having been solicited as the mean or the mode of the prior distribution of $\boldsymbol{\pi}$; i.e., $\mathbf{p}_{y|x_i}$ is treated as a noisy estimate of $\boldsymbol{\pi}$ which is subject to random error. The Bayes estimate of $H(\boldsymbol{\pi})$ under quadratic loss is the expected entropy given by:

$$\tilde{H}_i(\boldsymbol{\pi}|\boldsymbol{\alpha}, B) = \psi(B+1) - \frac{1}{B}\sum_{k=1}^{m} \alpha_{ik}\psi(\alpha_{ik}+1), \tag{30}$$

where $\psi(\alpha) = d\log\Gamma(\alpha)/d\alpha$ is the digamma function (Mazzuchi et al., 2008).

The information framework of Fig. 1 can be implemented in terms of (30). For $g(x_i) = 1/n, i = 1, \dots, n$, the consensus distribution is Dirichlet with parameter $\boldsymbol{\alpha}_c = \sum_{i=1}^{n} \boldsymbol{\alpha}_i/n$ and its kernel is the geometric mean of the individuals' kernels. The disagreement measure $K_c(\phi_{\pi|X} : \phi_c)$ can be computed using the KL function between two Dirichlet distributions (Ebrahimi et al., 2011).

7. Summary and discussion

We illustrated that the systematic application of the information theory to the mixture model proposed by Wallis (2005) for combining density forecasts provides a unifying framework for studying the notions of uncertainty, information, and disagreement of economic forecasters. In this framework, entropy measures uncertainty, the mutual information between the environment and economic variable measures the expected information that the environment provides for forecasting an economic variable, and the KL divergence provides a measure of disagreement between forecasters. The KL measure of disagreement is equivalent to the mutual information. This measure formalizes the idea that forecasters retain portions of the information which the environment provides for predicting the economic variable, along the lines of Sims's (Sims, 2003, 2006, 2011) theory of rational inattention.

We studied the dynamics of the evolution of uncertainty of the economic forecasters about the inflation rate over time using the SPF data. We illustrated application of the normalized entropy index for adjusting changes of the number of bins made in the SPF over two time periods. The adjustments corrected the distorted patterns of uncertainty measures due the changes in the number of bins. Using Bayesian hierarchical models, we explored the dynamics of the uncertainty about the inflation rate at the individual and aggregate levels. For each case, two sets of models are used, one for the uncertainty about the current year's inflation and one for the uncertainty about the following year's information. These models captured the seasonal variation of inflation uncertainty and the positive association between the uncertainty in the current quarter and uncertainty in the previous quarter. The seasonal effects are stronger for the uncertainty about the current year's inflation rate than about the following year's inflation rate. The uncertainty about inflation rate is positively correlated with the previous quarters' uncertainty, but the association diminishes quickly over time.

In addition to exploring the dynamics of the uncertainty about the inflation rate, we examined the relationships between inflation uncertainty and anticipated inflation. The results of individual-level models indicated that the relationship is plausible, but the quarterly aggregate model did not provide support for it. We also investigated the relationship between uncertainty and dispersion of the point forecasts at the aggregate level. The results were different for the uncertainty of the consensus distribution and the consensus uncertainty (the average of the entropies of individual forecast distributions). The results indicated that the relationship is plausible for the uncertainty of the consensus distribution. But we found no evidence for the relationship in the case of the consensus uncertainty.

We illustrated implementation of the information framework (Fig. 1) based on variance of the forecast distributions. The maximum entropy model based on the first two moments is normal distribution. Applying this method to the SPF data, we found that the entropy measures based on the variance are highly correlated with the entropy of the forecast distribution, however, the percentage of forecasters that they ranked equally decreased with the forecast horizon. Some researchers have questioned appropriateness of entropy for certain applications because the entropy is location invariant (Sofas, discussion of Bernardo, 1979b; Wallis, private communication). However, (25) reveals that, under certain conditions, the difference between two entropies can be expressed as a KL divergence which captures the difference between the location parameters such as the first term in (24).

We addressed the implementation of the information framework (Fig. 1) when the probability distribution of the economic variable is unknown and consequently quantities calculated from the solicited probabilities cannot be taken by their face values. We used Dirichlet models for the probability vector induced by the unknown distribution of the economic variable. In this approach, we used the vector of probabilities solicited from each forecaster as the mode of a Dirichlet distribution. The degree of belief parameter of the Dirichlet model, B, served as the tuning parameter for taking the inaccuracy of the solicited probabilities into account; e.g., in a classroom experiment, a smaller B is needed as compared with the respondents of the SPF survey.

The information framework (Fig. 1) can also be implemented when the forecasters are a set of econometrics models; e.g., Shoja (2015) considered the diffusion index models where the environment X is defined by the model space consisting of a set of linear models constructed by combinations of principle components of the regression matrix. The Bayesian posterior predictive distribution of each model is an economic forecaster, and the consensus distribution is found by Bayesian model averaging.

Finally, in addition to the proposed framework, the information theory can provide further insights about issues discussed in the literature on the economic forecasters; e.g., Wallis (2005) draws attention to some issues using a scenario where for forecasting an economic variable Y_t generated from an $AR(2)$ process, one uses $AR(1)$ and one uses y_{t-2}. Wallis points out that the consensus forecast model uses the same information set as the $AR(2)$ but inefficiently. The information measures of worth of observations (Pourahmadi and Soofi, 2000) can be used to assess the loss of information due to the missing lags in the two forecast models. Extension of these measures for mixture distribution can provide insights about assessing the inefficiency of the consensus forecast model and other issues raised by Wallis (2005).

Appendix

Disagreement based on Rényi information divergence.

The expression for Rényi information divergence between two normal distributions $f_i = N(\mu_i, \sigma_i^2),\ i = 1, 2$ can be written in the following form:

$$K_r(f_1 : f_2) = \frac{r(\mu_1 - \mu_2)^2}{2\sigma_r^2} + \frac{1}{2(r-1)} \log \frac{\sigma_2^2}{\sigma_r^2} - \frac{1}{2} \log \frac{\sigma_1^2}{\sigma_2^2},$$

which is defined when $\sigma_r^2 = (1-r)\sigma_1^2 + r\sigma_2^2 > 0$. In Example 2, $f_1 = f_{y|x}$ and $f_2 = f_y$. Noting that $\sigma_{y|x}^2 = (1-\rho^2)\sigma_y^2$, we have $\sigma_r^2 = [1 + (r-1)\rho^2]\sigma_y^2 > 0$ which holds for all $r > 0$. Then

$$K_r(f_{y|X} : f_y) = \frac{r(\alpha + \beta X - \mu_y)^2}{2[1 + (r-1)\rho^2]\sigma_y^2} + \frac{1}{2(r-1)} \log \frac{1}{[1 + (r-1)\rho^2} - \frac{1}{2} \log(1 - \rho^2).$$

The last term is the mutual information (17). Letting $\alpha = \mu_y - \beta\mu_x$, $\beta = \rho\frac{\sigma_y}{\sigma_x}$, and taking the expectation $E[K_r(f_{y|X} : f_y)]$, we obtain (18).

Proof of Proposition 1.
The ME model subject to the moment constraints (27) is given by:

$$f_c^*(y) = C \exp\left(-\sum_{j=1}^{J} \lambda_{c,j} y^j\right), \tag{31}$$

where $\lambda_{c,1}, \ldots, \lambda_{c,J}$ are the Lagrange multipliers for constraints (26) and $C = C(\lambda_{c_1}, \ldots, \lambda_{c,J})$ is the normalizing factor. (The ME model can be a probability vector or a continuous density). Breaking the log-ratio in $K(f_{y|x}^* : f_a^*)$, we have:

$$\begin{aligned} E_g[K(f_{y|X}^* : f_c^*)] &= E_g\left[-H(f_{y|X}^*) - E_{y|x}[\log f_c^*(Y)]\right] \\ &= -E_g\left[H(f_{y|X}^*)\right] - \log C + E_g\left[\sum_{j=1}^{J} \lambda_{c,j} E_{y|x}(Y^j)\right] \end{aligned}$$

$$= -E_g\left[H(f^*_{y|X})\right] - \log C + \sum_{j=1}^{r} \lambda_{c,j} E_g\left[E_c(Y^j)\right]$$

$$= -E_g\left[H(f^*_{y|X})\right] - \log C + \sum_{j=1}^{r} \lambda_{c,j} \mu_{c,j}.$$

The second equality is obtained using (31) and the last equality is from (27). The last result is found by noting that

$$H(f^*_c) = -\log C + \sum_{j=1}^{J} \lambda_{c,j} \mu_{c,j}.$$

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